

# Functional Requirements

When it comes to functional requirements basically it defines the specific features and behaviors of the software they represent the tasks a user must be able to complete to consider the system "working" so simply its like if the functional requirements answers (what does the system do)

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## User-Side Functions (Customer End)

### 1. User Registration and Authentication

The system shall allow users to create accounts and log in securely using credentials such as email and password this helps a lot as it gives each stakeholder which were emphasized earlier some sort of a unique id according to the system which helps a lot with the delivery system and also helps a lot during constructing the system especially in the database phase

### 2. Browse Restaurants and Menus

The system shall display a list of restaurants along with their menus, categories, and details this comes as the second step from the user perspective after the user signing in to the system just like ongoing food delivery systems (taalabat, etc)

### 3. Place Orders

The system shall allow customers to select food items, add them to a cart, and place orders

### 4. Special Requests

The system shall allow the user to add unique comments on the order example (user: asks for a burger but with no pickles) or other request like (the user asks the delivery not to knock on the door but send a notification via contact method)

### 5. Payment Processing

The system shall support multiple payment methods such as cash on delivery and online payment

### 6. Real-Time Order Tracking

The system shall provide real-time updates on the order status and its a pretty important feature as it helps with the users experience with the system

## **7. Notification System**

The system shall send notifications to users regarding order updates and delivery status

## **8. Rating & Review System**

Customers must be able to submit numerical ratings and text reviews for both the restaurant's food quality and the courier's service

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## **Restaurant & System-Side Functions**

All the previous functions are regarding the first end of the system which is the targeted users or the customer the upcoming part is regarding the other end which is the restaurants

### **8. Order Management for Restaurants**

The system shall allow restaurants to receive, view, and manage incoming orders this helps a lot with the process of sorting out the protocol of the restaurant for example whether its a fcfs which stands for (first come first serve) so simply it indicates that the first order to be placed is the first to be finished or another technique such as priority queue where they try to prioritize orders that require minimum amount of time to be first

### **9. Update Order Status**

The system shall allow restaurants to update the status of orders this comes mainly to ease up the process for the delivery man to know whether the order is ready to be picked up or its still not done

### **10. Delivery Assignment**

The system shall automatically assign delivery personnel to orders this increases the efficiency of the delivering process as it chooses specific employees to pick up certain orders

### **11. Order History**

The system shall store order history and provide basic analytics this helps both the user and system as it gives the system a genuine idea about the user favourite type of foods so it can recommend specific restaurants for the user to improve his experience as well as it helps the user to track his order history

### **12. Transaction, Refund Processing & Complaints**

The system must facilitate secure payment transfers between the customer and the

restaurant, while also allowing admins to process partial or full refunds in case of disputes

### **13. Delivery Tracking and Updates**

The system shall allow delivery personnel to update delivery status this is different from the real time order tracking as the other works on tracking the current location of the order while this one only shows whether the order was picked up from the restaurant or not

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As the functional requirements comes in two different way in user and system they both must be synchronized to ensure the efficiency of the system

# Non-Functional Requirements

Non-Functional Requirements define the quality attributes of the system. They focus on how the system performs its functions rather than what functions it provides. These requirements ensure that the system is efficient, secure, reliable, and user-friendly so it's like it answers **(how does the system performs)**

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## 1. Performance

The system should respond to user requests within a short time and handle multiple users simultaneously

## 2. Usability

The system should provide a simple and intuitive user interface

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## Security & Privacy

- **Data Encryption**

All sensitive data, including user passwords and payment tokens, must be encrypted at rest and in transit using AES-256 or higher

- **PCI Compliance**

The system must comply with PCI-DSS standards to ensure secure handling of credit card information

- **Privacy**

The system must comply with local data protection laws (like GDPR or CCPA), allowing users to request data deletion

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## 3. Availability & Maintenance

The system should be available 24/7 with minimal downtime

## 4. Maintenance

The system should be up to date always and fault tolerant while having a back up

## 5. Scalability

The system should support increasing numbers of users and orders

## 6. Reliability & Robustness

- **Fault Tolerance:** If the third-party GPS tracking service fails, the system should fail gracefully by showing the last known location rather than crashing the entire app